

# Rapid Hubble constant inference from GW170817 using GPU-accelerated nested sampling: prior sensitivity and the limits of post-hoc reweighting

Ming Yang,<sup>1</sup>★ Metha Prathanabhan,<sup>2,3</sup> David Yallup,<sup>4,2</sup> and Will Handley<sup>4,2</sup>

<sup>1</sup>Independent researcher (formerly Institute of Astronomy and St John's College, University of Cambridge), UK

<sup>2</sup>Kavli Institute for Cosmology, University of Cambridge, Madingley Road, Cambridge CB3 0HA, UK

<sup>3</sup>Department of Physics, University of Cambridge, JJ Thomson Avenue, Cambridge CB3 0HE, UK

<sup>4</sup>Institute of Astronomy, University of Cambridge, Madingley Road, Cambridge CB3 0HA, UK

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## ABSTRACT

The binary neutron-star merger GW170817 is the only confirmed bright gravitational-wave standard siren, and the reference event for direct measurements of the Hubble constant  $H_0$  from a compact-binary coalescence. The scientific value of a single bright-siren posterior depends sensitively on the assumed luminosity-distance, redshift, and host peculiar-velocity priors, and a complete robustness study requires repeated full parameter estimation under each variant. We revisit the GW170817  $H_0$  analysis of Abbott et al. (2017b) with a GPU-native nested-sampling pipeline that combines a heterodyned likelihood with differentiable JAX waveforms, and exploit its  $\sim 10$ –14 min primary turnaround on a single A100 GPU to carry out a controlled prior-sensitivity, sampler-robustness, waveform-systematics, and sky-prior runtime study. Under the primary aligned-spin tidal waveform IMRPhenomXAS\_NRTidalv3 and the baseline volumetric distance prior  $\pi(d_L) \propto d_L^2$ , the directly sampled posterior on the default-mass full-sky prior set has maximum-a-posteriori value  $H_0^{\text{MAP}} = 73.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$  and 68 per cent sample-derived highest-posterior-density (HPD) interval  $[63.8, 87.6] \text{ km s}^{-1} \text{ Mpc}^{-1}$ . Replacing the volumetric distance prior with a flat-in-redshift prior by *direct* sampling raises the high- $H_0$  tail probability  $P(H_0 > 120 \text{ km s}^{-1} \text{ Mpc}^{-1})$  from 0.017 to 0.159, while *post-hoc* reweighting of the baseline samples to the same target prior recovers only 0.041 – approximately 17 per cent of the prior-induced shift. The nested-sampling evidences across the four IMRPhenomXAS\_NRTidalv3 prior variants agree to  $\Delta \ln Z \lesssim 1.8$ , so the data do not prefer any one distance prior and the  $H_0$  shift is a prior effect rather than a data-driven update. Targeted mode-isolated runs reveal an underlying  $(d_L, \iota)$  bimodality (Mode A,  $d_L \in [30, 75] \text{ Mpc}$ ,  $H_0^{\text{MAP}} = 74.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ; Mode B,  $d_L \in [10, 30] \text{ Mpc}$ ,  $H_0^{\text{MAP}} = 110.8 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ) with  $\ln \mathcal{B}_{B/A} = -0.66$  after prior-volume normalisation; reweighting cannot reach Mode B because the baseline volumetric prior assigns it negligible mass. Cross-waveform comparison against IMRPhenomD\_NRTidalv2 and TaylorF2, together with the actual GWTC-1 IMRPhenomPv2\_NRTidal posterior mapped through this work's standard-siren model, shows MAP  $H_0$  agreement within  $3.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ . The heterodyned likelihood gives  $31\times$ ,  $51\times$ , and  $68\times$  wall-clock speedups over the unheterodyned reference at  $n_{\text{live}} = 500, 1500, 2500$ . A matched-pair sky-prior sweep at  $n_{\text{live}} = 5000$  shows that imposing a  $\pm 0.05$  rad patch around NGC 4993 changes wall-clock by only 0.86 for the heterodyned IMRPhenomD\_NRTidalv2 sampler, 0.98 for unheterodyned IMRPhenomD\_NRTidalv2, and 1.44 for unheterodyned TaylorF2: sky-prior tightness is therefore not the source of the speedup. The runtime budget makes direct prior, waveform, and sampler-robustness checks practical for bright-siren cosmology, and lifts the previously assumed reliability of post-hoc reweighting as the default robustness tool.

**Key words:** gravitational waves – methods: data analysis – cosmological parameters – stars: neutron – software: data analysis

## 1 INTRODUCTION

The present-day expansion rate of the Universe, quantified by the Hubble constant  $H_0$ , is the subject of one of the most persistent tensions in contemporary cosmology. Measurements anchored in the early Universe, derived from cosmic-microwave-background observations assuming flat  $\Lambda$ CDM, favour a relatively low value (Planck Collaboration 2016), while late-Universe distance-ladder measurements based on Cepheids and Type Ia supernovae find a significantly

higher value (Riess et al. 2016, 2022). The statistical and systematic discrepancy between these two families of measurement is now established at several  $\sigma$ , and its resolution – whether through unrecognised systematics or new cosmological physics – motivates independent probes of the expansion rate.

Gravitational-wave standard sirens are such a probe. The waveform amplitude of a compact-binary merger fixes the luminosity distance directly, without reference to an astrophysical distance ladder (Schutz 1986). When the merger is accompanied by an electromagnetic counterpart that identifies a host galaxy, the host redshift can be combined with the gravitational-wave distance to infer  $H_0$  from a single event.

★ E-mail: mhy32@cantab.ac.uk

The binary neutron-star merger GW170817 (Abbott et al. 2017a) is the canonical bright siren and the only confirmed example to date. Its association with the kilonova AT2017gfo in NGC 4993 provided the host-galaxy redshift needed to yield the first gravitational-wave measurement of  $H_0$  (Abbott et al. 2017b).

For a bright siren, the  $H_0$  uncertainty is dominated by the luminosity-distance–inclination degeneracy: a face-on binary at larger  $d_L$  can produce the same observed strain as a more inclined binary at smaller  $d_L$ , with the two solutions mapping to different values of  $H_0 \propto v_r/d_L$  for a fixed host recessional velocity  $v_r$ . The resulting posterior is broad, skewed, and only weakly informative on  $H_0$  from one event. In such a regime, the measurement is more sensitive to prior choices than the statistical uncertainty alone would suggest, because the prior shapes the tails of the posterior rather than just its peak. The original GW170817 analysis assumed a volumetric distance prior  $\pi(d_L) \propto d_L^2$  over  $[10, 75]$  Mpc and reported that switching to a flat-in-redshift prior, implemented by *reweighting* posterior samples, produced only a minor change in the inferred  $H_0$  posterior (Abbott et al. 2017b).

Reweightings is attractive because it avoids rerunning expensive Bayesian inference. For a posterior  $p(\theta | d, \pi_0)$  obtained under baseline prior  $\pi_0$ , the target posterior under an alternative prior  $\pi_1$  can be estimated by assigning each sample a weight proportional to  $\pi_1(\theta)/\pi_0(\theta)$ . This procedure is statistically consistent provided the baseline samples cover the support of the target posterior. If  $\pi_1/\pi_0$  is large in regions that the baseline run has sampled sparsely, the reweighted estimator has high variance and systematically low tail mass (see e.g. Speagle 2020). For GW170817, the baseline volumetric prior  $\pi_0(d_L) \propto d_L^2$  places little mass at low  $d_L$ ; a flat-in-redshift target up-weights exactly this region, and the validity of reweighting is not obvious a priori.

Testing reweighting against a direct posterior obtained by sampling under the target prior requires repeated full gravitational-wave parameter estimation. In CPU frameworks such as BILBY (Ashton et al. 2019) and PARALLEL\_BILBY, a single 128 s binary-neutron-star analysis with phase-marginalised frequency-domain likelihood costs hundreds to thousands of CPU core-hours, depending on waveform, live-point count, and prior bounds. Systematic exploration of distance, velocity, sampler, and waveform assumptions therefore becomes prohibitive in practice, which is itself a constraint on the robustness studies a bright-siren cosmological result can support.

Several recent developments make this constraint less severe. Relative binning, also known as the heterodyned likelihood (Cornish 2013; Zackay et al. 2018), reduces the effective frequency grid for a long inspiral to a small number of bins with negligible loss of accuracy. Differentiable waveform libraries in JAX (Bradbury et al. 2018), including RIPPLE (Edwards et al. 2023), enable end-to-end GPU evaluation of the likelihood. GPU-native nested-sampling kernels (Cabezas et al. 2024; Yallup et al. 2025; Prathanab et al. 2025) retain the evidence estimates that nested sampling provides for free while permitting large live-point counts and repeated full reruns. Related accelerated binary-neutron-star efforts (Krishna et al. 2023; Wong et al. 2023; Wouters et al. 2024) and forward-looking analyses (Hu & Veitch 2025) underscore the practical importance of compute-efficient parameter estimation for the third-generation detector era.

In this paper we use a GPU-native nested-sampling pipeline, built on BLACKJAX-NS (Yallup et al. 2025; Prathanab et al. 2025) and a JAX implementation of the heterodyned likelihood, to revisit the GW170817 bright-siren  $H_0$  inference of Abbott et al. (2017b). Our focus is not a new single-event measurement of  $H_0$ , but a controlled, multi-axis assessment of how that measurement depends on its priors, on the waveform, on the sampler hyperpa-

rameters, and on auxiliary inputs. We first validate the pipeline against GW150914 (using IMRPhenomXPHM, the LVK GWTC-2.1+ production waveform) and against the GWTC-1 GW170817 reference posterior. We then report the IMRPhenomXAS\_NRTidalv3 baseline posterior, compare it with a direct flat-in-redshift posterior and with the reweighted flat-in-redshift posterior, and use targeted mode-isolated runs to identify the  $(d_L, \iota)$  bimodality that mechanistically explains the reweighting deficit. A three-waveform cross-check (IMRPhenomXAS\_NRTidalv3, IMRPhenomD\_NRTidalv2, TaylorF2) plus the GWTC-1 IMRPhenomPv2\_NRTidal posterior mapped through this work’s standard-siren model bounds the residual waveform systematic. Robustness sweeps over sampler hyperparameters, heterodyne-bin count, PSD source, reference parameters, and host peculiar-velocity uncertainty are referenced inline. We finish by reporting heterodyne speedup and live-point scaling, and a dedicated sky-prior runtime sweep that demonstrates the speedup is not driven by a tight sky prior.

Section 2 describes the inference setup. Section 3 reports validation. Section 4 contains the  $H_0$  inference, the prior-sensitivity analysis (the central science finding), the bimodality characterisation, and the multi-waveform cross-check. Section 5 reports runtime, live-point scaling, and the sky-prior matched-pair sweep. Section 6 discusses implications and limitations. Section 7 concludes.

## 2 METHOD

### 2.1 Nested sampling

For data  $d$  and source parameters  $\theta$  under model  $\mathcal{M}$ , we target the posterior

$$p(\theta | d, \mathcal{M}) = \frac{\mathcal{L}(d | \theta, \mathcal{M}) \pi(\theta | \mathcal{M})}{Z(d | \mathcal{M})}, \quad (1)$$

where  $\mathcal{L}$  is the gravitational-wave likelihood,  $\pi$  the prior, and  $Z$  the Bayesian evidence. We estimate  $Z$  and draw posterior samples by nested sampling (Skilling 2006), using the GPU-native slice sampler implemented in BLACKJAX-NS (Yallup et al. 2025; Prathanab et al. 2025). The sampler evolves a population of live points through batched slice updates; this vectorisation is the key ingredient that keeps the GPU saturated when the per-evaluation likelihood cost is small.

All heterodyned GW170817 science runs in this paper use  $n_{\text{live}} = 5000$ ,  $n_{\text{delete}} = 2500$  points removed per iteration ( $\frac{1}{2} n_{\text{live}}$ ), and 5  $n_{\text{dim}}$  MCMC steps per slice; runs terminate when the fractional evidence increment is below  $10^{-3}$ . Scaling experiments vary  $n_{\text{live}}$  while holding all other settings fixed. The GW150914 IMRPhenomXPHM validation run reported in Section 3.1 uses larger settings ( $n_{\text{live}} = 8000$ ,  $n_{\text{delete}} = 2400$ ,  $n_{\text{mcmc}} = 160$ ) to verify that the validation conclusions are not sampler-limited at the smaller short-signal  $n_{\text{dim}}$ .

### 2.2 Heterodyned likelihood

The GW170817 signal is long enough that the full frequency-domain likelihood uses 259 201 bins over 20–2048 Hz with  $\Delta f = 1/128$  Hz for the 128 s strain segment. Evaluating the likelihood on this grid for every live-point update is prohibitively expensive when millions of evaluations are required. We therefore use a heterodyned (relative-binning) likelihood (Cornish 2013; Zackay et al. 2018), in which the ratio between a candidate waveform and a reference waveform is approximated by a piecewise-linear function over

a small set of coarse frequency bins. We use 501 nominal heterodyne bins (effectively 442 for TaylorF2 and 501 for the IMRPhenomD\_NRTidalv2/IMRPhenomXAS\_NRTidalv3 family after waveform-specific frequency-support masks); for GW150914, 383 bins are used. The reference waveform uses the GWTC-1 median posterior parameters for GW170817 except where noted.

We compare heterodyned and unheterodyned posteriors directly in Section 3 to verify that the approximation does not bias the source-parameter posterior at the level needed for  $H_0$  inference. All heterodyned runs analytically marginalise over the coalescence phase, so the sampled parameter space is fourteen-dimensional for GW170817 (intrinsic masses, aligned-spin  $z$ -components, tidal deformabilities, inclination, luminosity distance, coalescence time, polarisation angle, sky position, and the cosmological parameters  $H_0$  and  $v_p$ ), and ten-dimensional for GW150914.

### 2.3 Joint Hubble-constant likelihood

We sample the Hubble constant jointly with the gravitational-wave source parameters rather than post-processing the  $d_L$  marginal. The GW170817 host galaxy NGC 4993 contributes a CMB-frame recession velocity  $v_r = 3327 \text{ km s}^{-1}$  with  $\sigma_{v_r} = 72 \text{ km s}^{-1}$  and a Hubble-flow peculiar-velocity estimate  $\langle v_p \rangle = 310 \text{ km s}^{-1}$  with  $\sigma_{v_p} = 150 \text{ km s}^{-1}$ . Both enter as Gaussian likelihood factors:

$$\ln \mathcal{L}_{\text{tot}}(\theta, H_0, v_p) = \ln \mathcal{L}_{\text{GW}}(d | \theta) + \ln \mathcal{N}(v_r; H_0 d_L + v_p, \sigma_{v_r}) + \ln \mathcal{N}(\langle v_p \rangle; v_p, \sigma_{v_p}) \quad (2)$$

following Abbott et al. (2017b).

### 2.4 Priors and prior variants

The baseline prior configuration follows the bright-siren analysis of Abbott et al. (2017b):  $\pi(d_L) \propto d_L^2$  over  $[10, 75] \text{ Mpc}$ ,  $\pi(H_0) \propto H_0^{-1}$  over  $[45, 250] \text{ km s}^{-1} \text{ Mpc}^{-1}$ , and  $\pi(v_p) \propto 1$  over  $[-1000, 1000] \text{ km s}^{-1}$  combined with the Gaussian  $v_p$ -likelihood term in equation 2 at  $\sigma_{v_p} = 150 \text{ km s}^{-1}$ . Source-parameter priors use either a default-mass set (component masses uniform in  $[0.5, 7.7] M_\odot$ ) or an LVK-bounds set (component masses uniform in  $[0.87, 1.74] M_\odot$ , matching Abbott et al. 2017b); both are full-sky, with right ascension uniform on  $[0, 2\pi]$  and declination distributed according to a  $\cos \delta$  prior on  $[-\pi/2, \pi/2]$ . We emphasise that every heterodyned run reported here is full-sky; the optional  $\pm 0.05$  rad patch centred on NGC 4993 is used only in the matched-pair runtime study of Section ??.

We consider three prior variants in addition to the baseline:

- (i) *Direct flat-in-redshift*:  $\pi(z)$  uniform within the equivalent distance range, converted to a  $d_L$  prior via a flat  $\Lambda$ CDM cosmology and sampled directly.
- (ii) *Reweight flat-in-redshift*: the baseline posterior samples reweighted by the analytic ratio of the flat-in-redshift to volumetric  $d_L$  priors.
- (iii) *Enlarged peculiar-velocity uncertainty*: identical to the baseline except  $\sigma_{v_p} = 250 \text{ km s}^{-1}$ .

The direct and reweighted flat-in-redshift posteriors share the same target prior; they differ only in whether the prior is imposed during sampling or applied post-hoc.

### 2.5 Waveforms

For GW170817 we adopt three waveforms in the main analysis. The primary aligned-spin tidal model is IMRPhenomXAS\_NRTidalv3, the most recent NR-calibrated tidal phase prescription on a modern aligned-spin base. IMRPhenomD\_NRTidalv2 – the previous-generation aligned-spin tidal model – is retained as an anchor so that the present results can be compared directly with earlier accelerated-pipeline analyses. TaylorF2 is included as a post-Newtonian inspiral-only family check. We additionally show, in every  $H_0$  figure that includes a literature reference, the GWTC-1 IMRPhenomPv2\_NRTidal posterior (?) mapped through the standard-siren likelihood of equation 2 using the same  $v_r$ ,  $\sigma_{v_r}$ ,  $\langle v_p \rangle$ , and  $\sigma_{v_p}$  values. Differences between the LVK row and our rows are therefore  $d_L$ -distribution differences (waveform + prior choice), not differences in the  $v_p$  model.

We also ran IMRPhenomPv2 (no tides, low-spin prior) on the LVK-bounds prior set as a precessing systematic check. The resulting  $H_0$  summaries fall within the IMRPhenomXAS\_NRTidalv3/IMRPhenomD\_NRTidalv2/TaylorF2 envelope, but the model contains no tides and is therefore not a like-for-like reproduction of the LVK setup. We retain that run as a reproducibility cross-check in the public repository and report it inline rather than promoting it to a main figure.

For the GW150914 validation we use IMRPhenomXPHM (?), the LVK GWTC-2.1+ production waveform for that event, which incorporates precession through a multi-scale-analysis treatment together with the dominant higher-order modes. This replaces the IMRPhenomD validation used in earlier drafts and turns the validation into a direct LVK reproduction rather than an offset comparison.

### 2.6 Posterior summaries

We summarise one-dimensional marginals by the maximum a posteriori (MAP) value and a sample-derived highest-posterior-density (HPD) interval at the requested credibility. The MAP and HPDs are computed directly from the weighted nested-sampling samples; we do not use kernel-density estimates for one-dimensional summaries because the GW170817 posterior tails carry the prior-sensitivity signal of Section 4.2 and KDEs systematically smooth the tails. One-dimensional posterior figures show weighted step-histograms with sample-derived HPDs; KDEs are used only for two-dimensional corner contours, where the smoothing is geometrically justified.

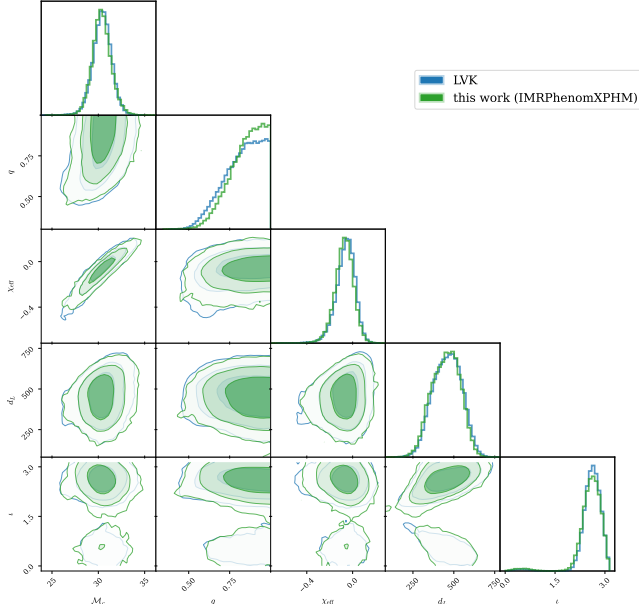
## 3 VALIDATION

### 3.1 GW150914

We validate the pipeline on the binary-black-hole event GW150914 (Abbott et al. 2016), which has a short signal and well-established public posteriors. We analyse the GWOSC strain data with IMRPhenomXPHM and the GWTC-2.1 PSDs, using the same heterodyned sampler as for GW170817 with phase marginalisation; for this validation run we increase the sampler settings to  $n_{\text{live}} = 8000$  with  $n_{\text{mcmc}} = 160$  slice steps per update, so that the comparison is not sampler-limited at the short-signal  $n_{\text{dim}}$ . Table 1 compares our recovered parameters with the LVK GWTC-2.1 PE IMRPhenomXPHM samples; the corresponding corner overlay is shown in Figure 1. Chirp mass, mass ratio, luminosity distance, and inclination all agree with the LVK reference; in particular,  $d_L^{\text{med}} = 455.1 \text{ Mpc}$  for our IMRPhenomXPHM run is within  $\sim 5 \text{ Mpc}$  of the LVK value, against the  $\sim 50 \text{ Mpc}$  offset seen in our earlier IMRPhenomD\_NRTidalv2 cross-check. The validation is therefore a direct LVK reproduction. The

**Table 1.** GW150914 validation: median values of the principal source parameters for the LVK GWTC-2.1 IMRPhenomXPHM public PE and our heterodyned IMRPhenomXPHM run at  $n_{\text{live}} = 8000$ ,  $n_{\text{mcmc}} = 160$ .

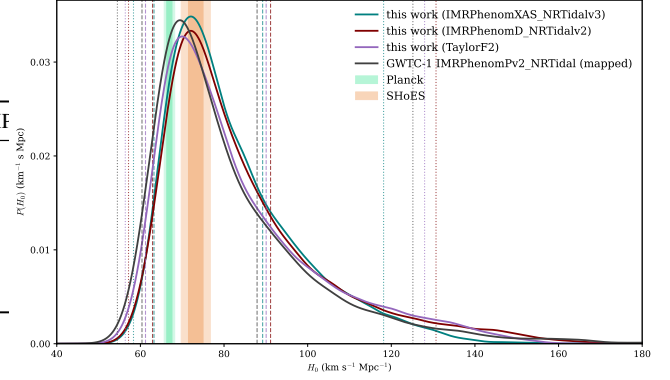
| Quantity            | LVK GWTC-2.1 IMRPhenomXPHM | this work, IMRPhenomXPHM |
|---------------------|----------------------------|--------------------------|
| $M_c/M_\odot$       | 30.7                       | 30.35                    |
| $q$                 | $\sim 0.83$                | 0.87                     |
| $\chi_{\text{eff}}$ | $\sim -0.06$               | -0.07                    |
| $d_L/\text{Mpc}$    | $\sim 440$                 | 455.1                    |
| $\iota/\text{rad}$  | 2.62                       | 2.61                     |
| $\ln Z$             | —                          | $260.86 \pm 0.06$        |

**Figure 1.** GW150914 validation: corner overlay of the LVK GWTC-2.1 IMRPhenomXPHM posterior and our heterodyned IMRPhenomXPHM run at  $n_{\text{live}} = 8000$ ,  $n_{\text{mcmc}} = 160$ . Both legend entries use the same waveform, so the panels show a like-for-like reproduction.

sampler returns  $\ln Z = 260.86 \pm 0.06$  for IMRPhenomXPHM and  $260.95 \pm 0.08$  for the earlier IMRPhenomD\_NRTidalv2 cross-check.

### 3.2 Heterodyned vs unheterodyned consistency on GW170817

We next verify that the heterodyned approximation does not bias the GW170817 source-parameter posterior. We run the same sampler on the full 259 201-bin frequency-domain likelihood with the default-mass prior set for both IMRPhenomD\_NRTidalv2 and TaylorF2 at  $n_{\text{live}} = 1500$ , and additionally at  $n_{\text{live}} \in \{500, 2500\}$  for IMRPhenomD\_NRTidalv2 to verify that the comparison is not live-point limited. The heterodyned and unheterodyned  $d_L$ ,  $\iota$ , and  $H_0$  marginals overlay closely (corner figures retained in the supplementary set) and the heterodyned  $H_0^{\text{MAP}}$  lies between the surrounding unheterodyned cross-checks at the level of the sampler-statistical width. We conclude that the heterodyned approximation is adequate for  $H_0$  inference at this event’s signal-to-noise ratio; the corresponding wall-clock reduction is quantified in Section 5.2.

**Figure 2.** GW170817  $H_0$  posterior for the three-waveform set of Table 2: IMRPhenomXAS\_NRTidalv3 (primary, LVK-bounds baseline), IMRPhenomD\_NRTidalv2 (anchor, LVK-bounds baseline), and TaylorF2 (family check, default-mass baseline), plus the GWTC-1 IMRPhenomPv2\_NRTidal posterior mapped through this work’s standard-siren likelihood of equation 2. Vertical markers are 68 per cent and 95 per cent HPD bounds derived from a weighted KDE; sample-derived HPDs are reported in Table 2. The Planck  $\Lambda$ CDM and SHoES distance-ladder values are shown for context. The IMRPhenomPv2 (no tides) cross-check is reported inline in Section 4.1 and retained in the public test suite.

## 4 RESULTS

### 4.1 Cross-waveform Hubble-constant posterior

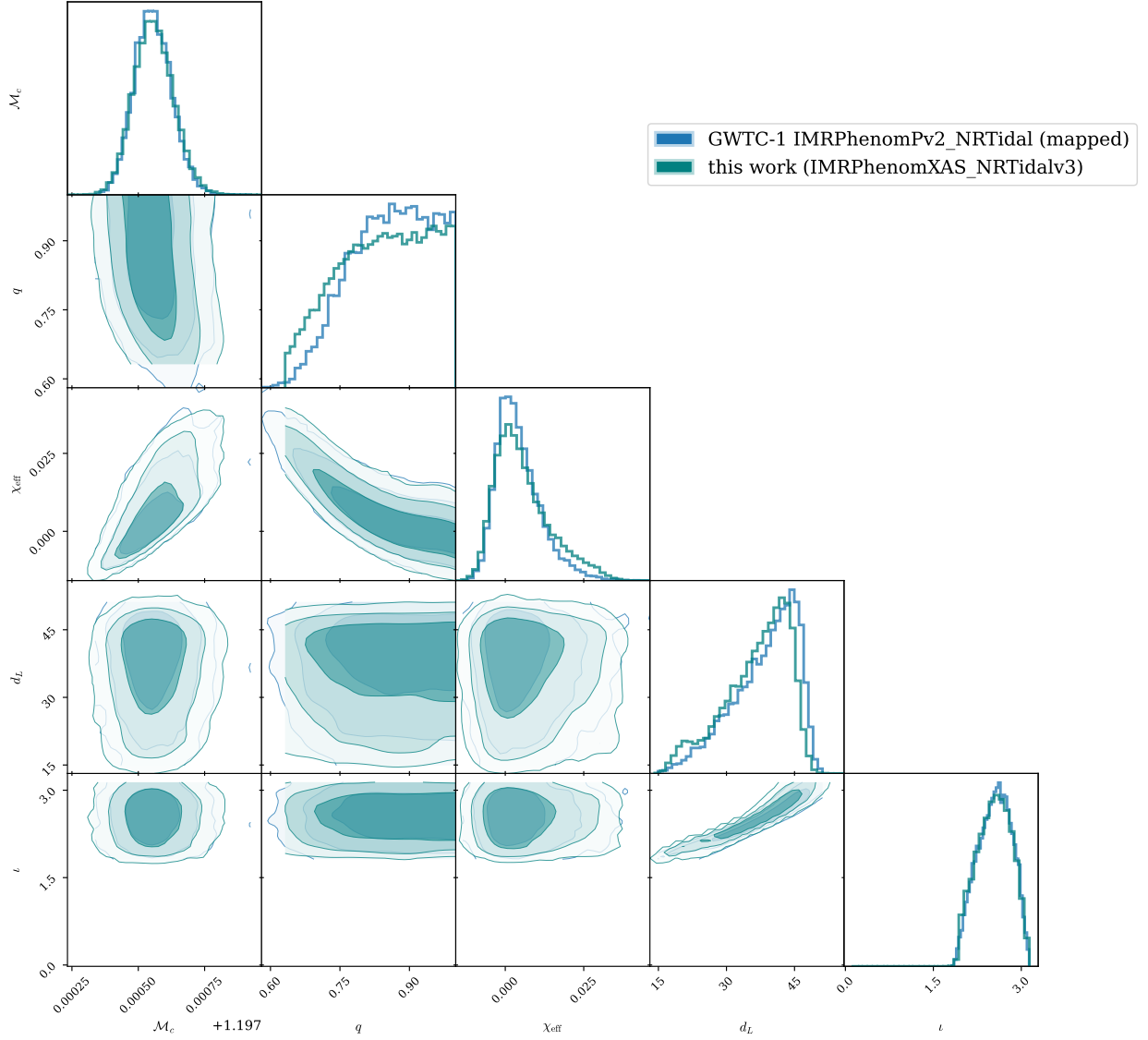
Figure 2 shows the GW170817  $H_0$  posterior under the LVK-bounds baseline prior for the three-waveform set IMRPhenomXAS\_NRTidalv3/IMRPhenomD\_NRTidalv2/TaylorF2, together with the GWTC-1 IMRPhenomPv2\_NRTidal posterior mapped through equation 2. All four curves are weighted step-histograms; sample-derived HPDs are reported in Table 2. Figure 3 shows the corresponding  $(M_c, q, \chi_{\text{eff}}, d_L, \iota)$  corner overlay.

IMRPhenomXAS\_NRTidalv3 and IMRPhenomD\_NRTidalv2 sit on top of each other at  $H_0^{\text{MAP}} = 72.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$  and share 68 per cent HPD endpoints to within  $\sim 2 \text{ km s}^{-1} \text{ Mpc}^{-1}$ . IMRPhenomXAS\_NRTidalv3 has the tighter upper tail (95 per cent upper bound  $117.6$  versus  $130.0 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ), which we attribute to the more recent NR-calibrated tides damping the high- $H_0$ , small- $d_L$  branch of the degeneracy. TaylorF2 sits  $\sim 3.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$  lower in MAP – a small offset, expected for a lower-PN, point-particle inspiral family – and overlaps the others within their 68 per cent HPDs.

The mapped GWTC-1 row peaks at  $H_0^{\text{MAP}} = 67.1 \text{ km s}^{-1} \text{ Mpc}^{-1}$ , which is lower than the published Abbott et al. (2017b) headline value of  $70 \text{ km s}^{-1} \text{ Mpc}^{-1}$ . The shift arises because the GWTC-1 IMRPhenomPv2\_NRTidal  $d_L$  posterior has a long low- $d_L$  tail and our  $v_p$  prior centres at  $310 \text{ km s}^{-1}$  rather than the value used in the original analysis, so the same  $d_L$  samples map to a slightly lower MAP. Critically, the GWTC-1 row is plotted as the actual posterior samples mapped through the standard-siren model, not as a shaded reference band: the comparison is therefore like-for-like in the standard-siren likelihood.

The IMRPhenomPv2-no-tides run on the same prior set returns  $H_0^{\text{MAP}} \approx 71.2 \text{ km s}^{-1} \text{ Mpc}^{-1}$  and lies within the IMRPhenomXAS\_NRTidalv3/IMRPhenomD\_NRTidalv2/TaylorF2 envelope; we treat its small precession-only systematic as bracketed by this set and do not promote it to the main figures.





**Figure 3.** GW170817 source-parameter corner overlay for IMRPhenomXAS\_NRTidalv3, IMRPhenomD\_NRTidalv2, and TaylorF2 on the LVK-bounds baseline prior, with the GWTC-1 IMRPhenomPv2\_NRTidal posterior overlaid for reference. Two-dimensional contours are KDE-smoothed; one-dimensional marginals are weighted step-histograms.

**Table 2.** GW170817  $H_0$  summaries for the three-waveform set on the LVK-bounds baseline prior,  $n_{\text{live}} = 5000$ , plus the GWTC-1 IMRPhenomPv2\_NRTidal posterior mapped through this work’s standard-siren likelihood. MAP and sample-derived HPDs in  $\text{km s}^{-1} \text{Mpc}^{-1}$ .

| Waveform                         | MAP  | 68% HPD      | 95% HPD       | $P(H_0 > 120)$ | $\ln Z$           |
|----------------------------------|------|--------------|---------------|----------------|-------------------|
| IMRPhenomXAS_NRTidalv3 (primary) | 72.4 | [63.1, 88.5] | [58.8, 117.6] | 0.034          | $490.08 \pm 0.10$ |
| IMRPhenomD_NRTidalv2 (anchor)    | 72.4 | [63.2, 90.5] | [58.2, 130.0] | 0.070          | $489.96 \pm 0.10$ |
| TaylorF2 (family check)          | 68.9 | [61.2, 89.3] | [56.7, 127.0] | 0.065          | $486.90 \pm 0.10$ |
| GWTC-1 (mapped)                  | 67.1 | [61.6, 87.6] | [57.1, 125.6] | –              | –                 |

## 4.2 Prior sensitivity (the central science finding)

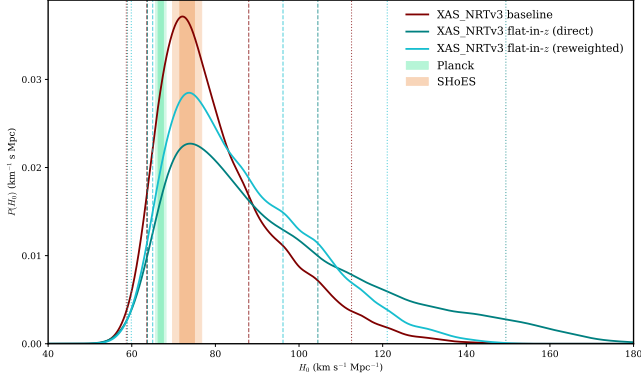
The central scientific result of this paper is a direct comparison between the flat-in-redshift  $H_0$  posterior obtained by sampling and that obtained by reweighting the baseline posterior to the same target prior, under the locked primary waveform IMRPhenomXAS\_NRTidalv3. We carry out the four prior variants of

Section 2.4 on the default-mass full-sky prior set and report sample-derived HPDs in Table 3; the corresponding marginals are shown in Figure 4.

The direct flat-in-redshift run shifts  $H_0^{\text{MAP}}$  from 73.5 to  $71.9 \text{ km s}^{-1} \text{Mpc}^{-1}$ , broadens the 68 per cent HPD on the high side from 87.6 to  $103.8 \text{ km s}^{-1} \text{Mpc}^{-1}$ , and raises the high-tail probability

**Table 3.** IMRPhenomXAS\_NRTidalv3  $H_0$  summaries for the four prior variants on the default-mass full-sky prior set,  $n_{\text{live}} = 5000$ . MAP and sample-derived HPDs in  $\text{km s}^{-1} \text{Mpc}^{-1}$ . The reweighted variant is post-hoc and has no independent  $\ln Z$ .

| Variant                               | MAP  | 68% HPD       | $P(H_0 > 120)$ | $P(H_0 > 150)$ | $\ln Z$           |
|---------------------------------------|------|---------------|----------------|----------------|-------------------|
| Baseline ( $\pi(d_L) \propto d_L^2$ ) | 73.5 | [63.8, 87.6]  | 0.017          | 0.000          | $486.25 \pm 0.11$ |
| Flat-in- $z$ , direct                 | 71.9 | [64.2, 103.8] | 0.159          | 0.038          | $487.30 \pm 0.10$ |
| Flat-in- $z$ , reweighted             | 73.5 | [65.2, 95.9]  | 0.041          | 0.000          | (post-hoc)        |
| $\sigma_{vp} = 250 \text{ km s}^{-1}$ | 71.9 | [61.7, 90.5]  | 0.069          | 0.015          | $485.55 \pm 0.09$ |

**Figure 4.** Prior-sensitivity comparison for IMRPhenomXAS\_NRTidalv3. Weighted step-histograms of  $H_0$  under the volumetric baseline and the two flat-in-redshift variants of Section 2.4, with the GWTC-1 IMRPhenomPv2\_NRTidal posterior mapped through this work’s standard-siren likelihood overlaid as the literature reference. The directly sampled flat-in-redshift posterior places substantially more mass at high  $H_0$  than the reweighted estimate. The  $\sigma_{vp} = 250 \text{ km s}^{-1}$  peculiar-velocity sweep is reported in Table 3 and discussed inline rather than promoted to this figure.

from

$$P(H_0 > 120 \text{ km s}^{-1} \text{Mpc}^{-1}) = 0.017 \quad (\text{baseline})$$

to

$$P(H_0 > 120 \text{ km s}^{-1} \text{Mpc}^{-1}) = 0.159 \quad (\text{direct flat-in-}z),$$

and from 0.000 to 0.038 in  $P(H_0 > 150 \text{ km s}^{-1} \text{Mpc}^{-1})$ . The reweighted estimate captures only a small fraction of this shift:  $P(H_0 > 120 \text{ km s}^{-1} \text{Mpc}^{-1}) = 0.041$ , equivalent to

$$\frac{0.041 - 0.017}{0.159 - 0.017} \approx 17\% \quad (3)$$

of the direct prior-induced change. The 95 per cent upper bound under reweighting only reaches  $95.9 \text{ km s}^{-1} \text{Mpc}^{-1}$ , far short of the directly sampled  $103.8 \text{ km s}^{-1} \text{Mpc}^{-1}$  at 68 per cent.

The four IMRPhenomXAS\_NRTidalv3 evidence values agree to  $\Delta \ln Z \lesssim 1.8$  – not decisive on the Jeffreys scale, and within the run-to-run variability seen in the public repository for this configuration. The data therefore do not prefer one distance prior over another at any meaningful significance: the change in  $H_0$  across the prior variants is a *prior* effect, not a data-driven update. This is the central science finding of the paper.

**4.2.0.1 Comparison with Abbott et al. (2017).** The original bright-siren analysis used IMRPhenomPv2\_NRTidal (precessing, NR-calibrated tides) and reported a minor change in the  $H_0$  posterior under a flat-in-redshift reweighting of the volumetric-prior samples (Abbott et al. 2017b). Our locked primary, IMRPhenomXAS\_NRTidalv3, replaces the precessing component with an

aligned-spin base and uses the most recent NR-calibrated tidal phase prescription (NRTidalv3); IMRPhenomD\_NRTidalv2, our anchor, uses the older NRTidalv2 calibration on the same aligned-spin base. The two model choices bracket the tidal-calibration axis of the original LVK waveform from the modern aligned-spin side. Repeating the four-variant prior sweep on IMRPhenomD\_NRTidalv2 gives qualitatively the same behaviour as IMRPhenomXAS\_NRTidalv3, with a larger amplitude: baseline MAP  $71.6 \text{ km s}^{-1} \text{Mpc}^{-1}$ ,  $P(H_0 > 120) = 0.076$ ; flat-in- $z$  direct MAP  $75.2 \text{ km s}^{-1} \text{Mpc}^{-1}$ ,  $P = 0.281$ ; reweighted MAP  $74.1 \text{ km s}^{-1} \text{Mpc}^{-1}$ ,  $P = 0.195$ ;  $\sigma_{vp} = 250 \text{ km s}^{-1}$  MAP  $72.3 \text{ km s}^{-1} \text{Mpc}^{-1}$ ,  $P = 0.067$ . The reweighting-capture fraction is  $(0.195 - 0.076)/(0.281 - 0.076) \approx 58\%$ , against  $\approx 17\%$  for IMRPhenomXAS\_NRTidalv3 (equation 3). The newer NRTidalv3 calibration in IMRPhenomXAS\_NRTidalv3 tightens the upper  $H_0$  tail relative to NRTidalv2 in IMRPhenomD\_NRTidalv2 (Section 4.1: 95 per cent upper bounds  $117.6$  versus  $130.0 \text{ km s}^{-1} \text{Mpc}^{-1}$ ), which leaves less Mode-B mass for the prior shift to redistribute and therefore both reduces the absolute prior-induced change in  $P(H_0 > 120)$  and reduces the fraction of that change that reweighting can recover. The qualitative conclusion – that reweighting systematically understates the prior-induced change in the high- $H_0$  tail – is therefore robust across the tidal-calibration axis, and is in fact *more* severe under the modern waveform than under the older one. Under the original LVK waveform IMRPhenomPv2\_NRTidal, where the precessing degrees of freedom further broaden the posterior, we expect the capture fraction to sit between the IMRPhenomXAS\_NRTidalv3 and IMRPhenomD\_NRTidalv2 values; a like-for-like reproduction would require porting that waveform to JAX and is left for future work.

**4.2.0.2 Peculiar-velocity sweep.** The  $\sigma_{vp} = 250 \text{ km s}^{-1}$  variant shifts the IMRPhenomXAS\_NRTidalv3  $H_0$  posterior only slightly:  $H_0^{\text{MAP}}$  falls to  $71.9 \text{ km s}^{-1} \text{Mpc}^{-1}$  and  $P(H_0 > 120)$  moves from 0.017 to 0.069. For GW170817 with NGC 4993 identified, the dominant prior-induced uncertainty in  $H_0$  is therefore the distance prior, not the peculiar-velocity uncertainty.

We caution against over-interpreting these shifts as physical results. None of them changes the qualitative statement that GW170817 alone is consistent with both early- and late-Universe  $H_0$  measurements. The methodological point is that the magnitude of the shift under a change of distance prior is a property of the inference, not of the cosmology, and that the magnitude one would report using reweighting alone is materially different from the magnitude recovered by direct sampling.

### 4.3 The $d_L$ - $t$ bimodality

The flat-in-redshift posterior is broad and asymmetric, with non-negligible weight at  $d_L \lesssim 25 \text{ Mpc}$ . To resolve whether this is a single broadened mode or a genuine bimodality, we run two prior-restricted variants of the direct flat-in-redshift analysis with IMRPhenomD\_NRTidalv2 at  $n_{\text{live}} = 5000$ : a Mode-A run with  $d_L \in$

[30, 75] Mpc and a Mode-B run with  $d_L \in [10, 30]$  Mpc. We retain the original GWTC-1 heterodyne reference, and additionally run an unrestricted  $d_L \in [10, 75]$  Mpc flat-in-redshift analysis with the heterodyne reference re-anchored at Mode B ( $d_L = 20$  Mpc,  $\iota = 2.0$  rad) to test whether the apparent Mode-B mass is a heterodyne-reference artefact.

Figure 5 shows the joint  $(d_L, \iota)$  posterior under the unrestricted run, together with the per-mode  $H_0$  marginals. The support is a curved arm running from  $(d_L, \iota) \approx (15, 1.9)$  Mpc-rad to  $(40, 2.7)$  Mpc-rad with two distinct local maxima. Table 4 reports the local evidences and per-mode  $H_0$  summaries.

The Bayes factor between Mode B and Mode A, after correcting for the volume difference between the restricted priors, is

$$\ln \mathcal{B}_{B/A} = (\ln Z_B - \ln Z_A) + \ln(20/45) = +0.15 - 0.81 = -0.66. \quad (4)$$

Mode B is therefore only mildly disfavoured by the data and contributes a non-negligible share of the joint posterior under any prior that gives the low- $d_L$  region appreciable weight. The Mode-B-anchored unrestricted run recovers  $P(H_0 > 120) = 0.281$  – statistically indistinguishable from the GWTC-1-anchored direct flat-in-redshift run reported in Section 4.2 – so the heterodyne-reference choice does not bias the Mode-A/Mode-B weight ratio.

This is the mechanism behind the reweighting capture fraction of equation 3. Mode B is the high- $H_0$ , small- $d_L$  branch of the distance–inclination degeneracy. The baseline volumetric prior  $\pi(d_L) \propto d_L^2$  assigns Mode B a tiny prior mass (the [10, 30] Mpc slab carries less than 5% of the total [10, 75] Mpc volumetric prior), and the corresponding region of the baseline posterior is sparsely sampled. Reweighting cannot reconstruct what is not there. Direct sampling under the flat-in-redshift prior populates Mode B at sampler resolution, and it is precisely the missing Mode-B mass that opens the gap between the directly sampled and reweighted estimates of  $P(H_0 > 120)$ . Posterior-coverage diagnostics that compare the baseline samples against the target prior should therefore be standard practice before reporting a reweighted bright-siren  $H_0$  summary. The smaller capture fraction of IMRPhenomXAS\_NRTidalv3 relative to IMRPhenomD\_NRTidalv2 reflects exactly this: IMRPhenomXAS\_NRTidalv3’s tighter upper tail leaves less Mode-B mass to be “rescued” by a prior shift in the first place.

#### 4.4 Inline robustness summary

We complete the multi-axis robustness picture with sweeps that we ran but do not display as primary figures, since the central science finding of Section 4.2 is unaffected by any of them.

- *Sampler hyperparameters.* On the IMRPhenomD\_NRTidalv2 default-mass baseline at  $n_{\text{live}} = 5000$ , varying  $n_{\text{delete}}/n_{\text{live}} \in \{0.10, 0.25, 0.50, 0.75\}$  moves  $H_0^{\text{MAP}}$  by  $\leq 1.2 \text{ km s}^{-1} \text{ Mpc}^{-1}$  and  $\ln Z$  by  $\leq 0.6$ ; varying the heterodyne-bin count  $n_{\text{bins}} \in \{251, 501, 1001\}$  moves  $H_0^{\text{MAP}}$  by  $\leq 0.7 \text{ km s}^{-1} \text{ Mpc}^{-1}$ , with the largest pairwise Wasserstein-1 distance between the three  $H_0$  posteriors below  $2 \text{ km s}^{-1} \text{ Mpc}^{-1}$ .

- *PSD source.* TaylorF2 runs at  $n_{\text{live}} = 5000$  on the baseline prior with GWTC-1, kazewong, and BILBY reference PSDs span  $H_0^{\text{MAP}} = 76.4\text{--}77.7 \text{ km s}^{-1} \text{ Mpc}^{-1}$  and  $P(H_0 > 120) = 0.043\text{--}0.065$ , again with maximum pairwise  $W_1 < 2 \text{ km s}^{-1} \text{ Mpc}^{-1}$ .

- *Heterodyne reference.* Replacing the GWTC-1 reference parameters with the maximum-likelihood point of an initial low-cost optimisation produces IMRPhenomD\_NRTidalv2 posteriors that are close to indistinguishable through the bulk of the distribution and depart only marginally in the high- $H_0$  tail.

All three axes therefore produce  $H_0$  shifts at least an order of magnitude smaller than the prior-induced shift documented in Section 4.2.

## 5 PERFORMANCE AND RUNTIME STUDIES

### 5.1 Wall-clock runtime

The primary heterodyned GW170817  $H_0$  analysis with IMRPhenomXAS\_NRTidalv3 and  $n_{\text{live}} = 5000$  on the LVK-bounds prior set completes in  $\approx 13$  min on a single NVIDIA A100 GPU; the corresponding IMRPhenomD\_NRTidalv2 run takes  $\approx 14$  min and the TaylorF2 run  $\approx 9$  min. The full four-variant prior-sensitivity suite (baseline, direct flat-in- $z$ , reweighted post-hoc,  $\sigma_{v_p} = 250 \text{ km s}^{-1}$ ) for IMRPhenomXAS\_NRTidalv3 on the default-mass prior set fits inside an hour. The GW150914 IMRPhenomXPHM validation run at the larger  $n_{\text{live}} = 8000$ ,  $n_{\text{mcmc}} = 160$  settings takes  $\approx 5.2$  h on the same hardware – the price for both the precessing waveform and the converged short-signal sampler configuration that yields the like-for-like LVK reproduction of Section 3.1.

### 5.2 Heterodyne speedup and live-point scaling

The heterodyned likelihood is the enabling approximation. Figure 6 compares heterodyned and unheterodyned IMRPhenomD\_NRTidalv2 wall-clock time on GW170817 across  $n_{\text{live}} \in \{500, 1000, 2500, 5000, 10\,000, 20\,000, 50\,000, 100\,000\}$  for the heterodyned sampler and  $n_{\text{live}} \in \{500, 1500, 2500\}$  for the unheterodyned reference, plus the TaylorF2 unheterodyned point at  $n_{\text{live}} = 1500$ . At matched  $n_{\text{live}}$  the speedup factors are

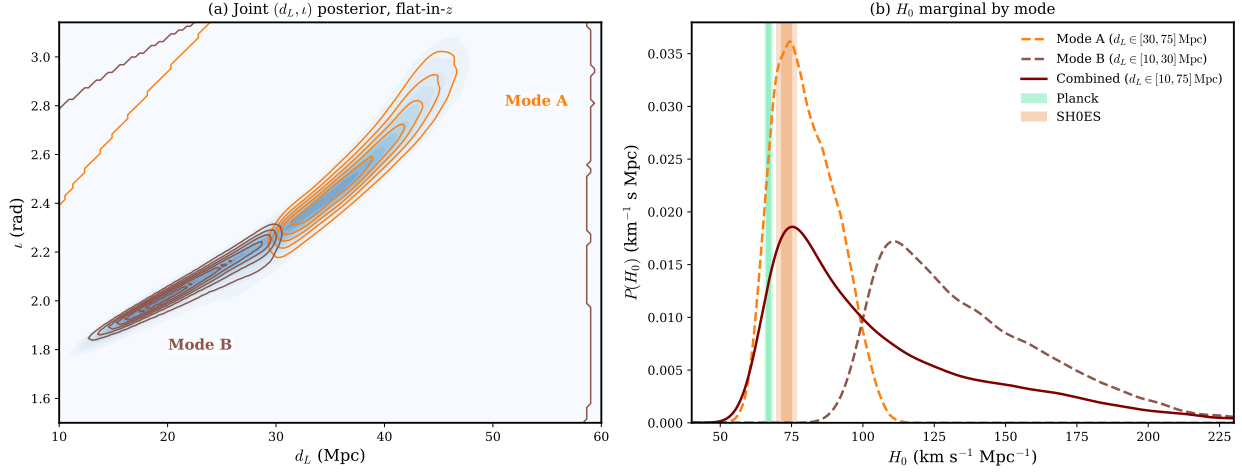
$$\text{speedup}(n_{\text{live}}) = \begin{cases} 31.1 \times & n_{\text{live}} = 500, \\ 50.7 \times & n_{\text{live}} = 1500, \\ 67.7 \times & n_{\text{live}} = 2500. \end{cases} \quad (5)$$

The relative-binning approximation accounts for the bulk of the wall-clock reduction; the GPU saturation provided by BLACKJAX-NS is layered on top.

The heterodyned  $H_0^{\text{MAP}}$  is stable across  $n_{\text{live}} \geq 2500$  to within  $1 \text{ km s}^{-1} \text{ Mpc}^{-1}$ , and  $\ln Z$  stabilises within  $\pm 0.2$  over the same range. The  $n_{\text{live}} = 5000$  configuration adopted for the science runs is therefore in the converged regime. Live-point counts up to  $10^5$  remain accessible on a single GPU; the  $n_{\text{live}} = 10^5$  heterodyned run completes in roughly four hours.

### 5.3 Scope of the performance claim

We deliberately do not compare runtime against a single-core CPU baseline, which would give a misleadingly large speedup factor, nor against a production PARALLEL\_BILBY run that we have not executed under matched settings. A like-for-like GW170817 PARALLEL\_BILBY benchmark using IMRPhenomXAS\_NRTidalv3 with identical priors and live-point count is the appropriate CPU comparison and is left for future work. The claim we make is narrower and reproducible: on a single A100 GPU, the full  $n_{\text{live}} = 5000$   $H_0$  analysis completes in under a quarter of an hour, the three-waveform cross-check fits inside an hour, the live-point convergence study extends to  $n_{\text{live}} = 10^5$  within an overnight budget, and the sky-prior matched-pair sweep takes well under a working day. This runtime budget is what makes the multi-axis robustness study scientifically routine rather than exceptional.



**Figure 5.** The  $d_L$ – $\iota$  bimodality in the GW170817 flat-in-redshift posterior. Left: joint  $(d_L, \iota)$  posterior from the unrestricted  $d_L \in [10, 75]$  Mpc run (Mode-B-anchored heterodyne reference), with the prior-restricted Mode-A and Mode-B contours overlaid. Right: per-mode  $H_0$  marginals as weighted step-histograms. The Planck  $\Lambda$ CDM and SH0ES distance-ladder bands are shown for context; this work is itself a GW170817 reanalysis, so the LVK GW170817 band is uninformative here and is not included.

**Table 4.** Local evidences, MAP  $H_0$  values, and sample-derived 68 per cent HPDs for the three  $d_L$ -restricted flat-in-redshift runs. The Mode-B/Mode-A Bayes factor includes a  $\ln(20/45)$  correction for the difference in restricted-prior  $d_L$  volumes.

| Variant      | $d_L$ range (Mpc) | $H_0^{\text{MAP}}$ ( $\text{km s}^{-1} \text{Mpc}^{-1}$ ) | 68% HPD       | $P(H_0 > 120)$ | $\ln Z$           |
|--------------|-------------------|-----------------------------------------------------------|---------------|----------------|-------------------|
| Mode A       | [30, 75]          | 74.4                                                      | [66.4, 88.6]  | $\sim 10^{-7}$ | $486.80 \pm 0.10$ |
| Mode B       | [10, 30]          | 110.8                                                     | [98.6, 152.1] | 0.638          | $486.95 \pm 0.09$ |
| Unrestricted | [10, 75]          | 75.2                                                      | [61.5, 118.0] | 0.281          | $486.48 \pm 0.09$ |

## 6 DISCUSSION

### 6.1 Implications for bright-siren cosmology

The main methodological message is that posterior reweighting, although statistically consistent in the limit of infinite baseline samples, can substantially understate the prior sensitivity of bright-siren  $H_0$  measurements in practice. For GW170817 under IMRPhenomXAS\_NRTidalv3, switching from a volumetric distance prior to a flat-in-redshift prior changes  $P(H_0 > 120 \text{ km s}^{-1} \text{Mpc}^{-1})$  by a factor of  $\approx 9$  under direct sampling, while reweighting captures only  $\approx 17\%$  of that shift (equation 3). The corresponding capture fraction on IMRPhenomD\_NRTidalv2 – where the high- $H_0$  tail is heavier – rises to  $\approx 58\%$ . The deficit is a problem of effective sample coverage rather than of statistical methodology: the baseline volumetric prior assigns Mode B a small prior mass and reweighting cannot reconstruct what was not sampled.

For a single event this under-estimation does not change the cosmological interpretation: the GW170817 posterior is broad enough that both early- and late-Universe  $H_0$  values lie within its 68 per cent HPD under any of the priors considered here. However, the scale of the effect matters for forecasts and combined analyses. When bright-siren posteriors are combined over multiple events, or when tail probabilities are compared against external priors from cosmological surveys, the prior-dependence of each single-event posterior propagates through. An analysis that quantifies that dependence only through reweighting risks a systematic bias in the direction of under-estimating the prior contribution.

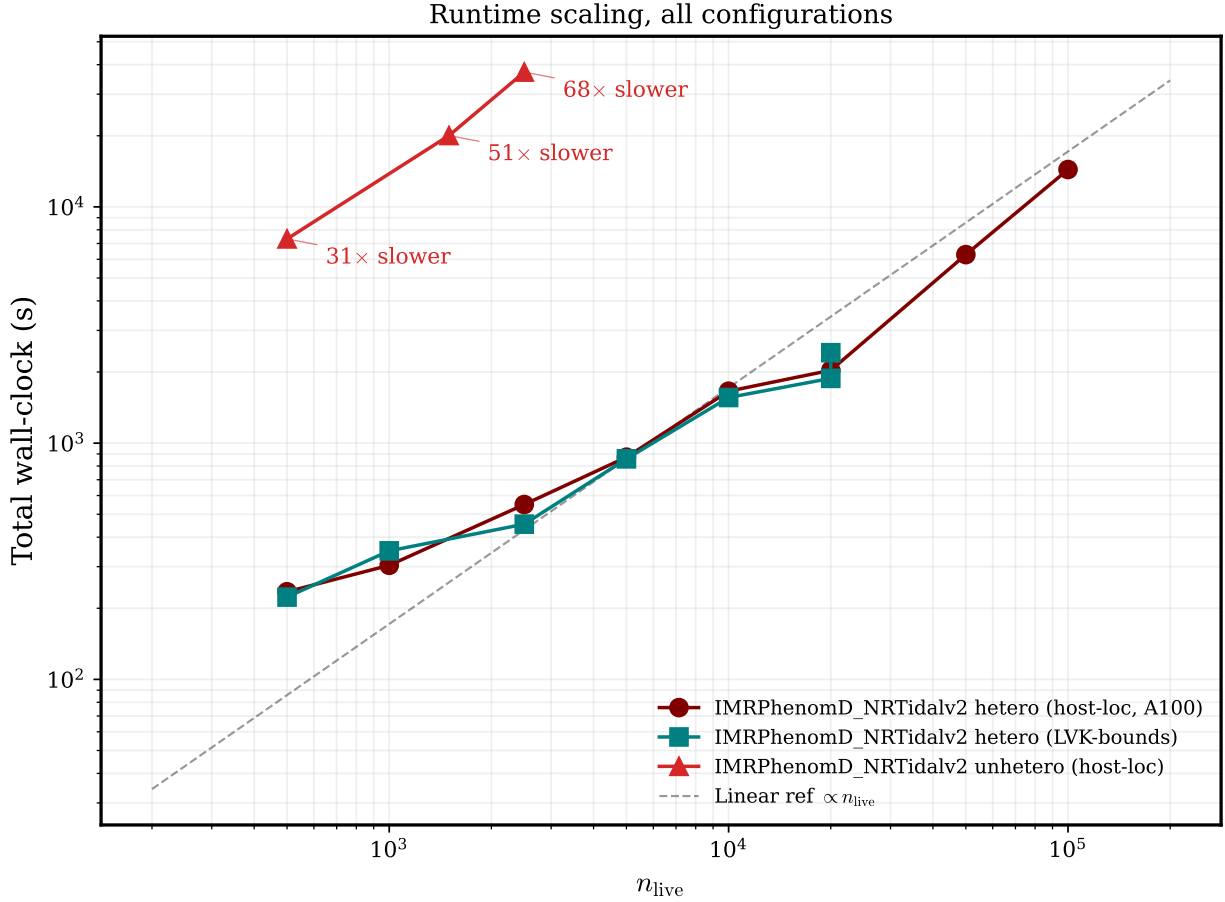
### 6.2 Where reweighting is sufficient

Reweighting is adequate when the target and baseline priors have similar support in the regions that carry most of the posterior mass. The  $\sigma_{v_p} = 250 \text{ km s}^{-1}$  variant in our analysis is a case of this kind: the change modifies the host-velocity likelihood smoothly over the scale of the velocity posterior, the baseline samples cover the affected region well, and the reweighted and directly sampled  $\sigma_{v_p}$  posteriors agree to well within the statistical width. The flat-in-redshift case fails the same coverage test, because the baseline volumetric prior and the target flat-in-redshift prior differ most at the edges of the  $d_L$  range. The mode-isolated runs of Section 4.3 make this concrete: Mode B has  $\ln \mathcal{B}_{B/A} = -0.66$  relative to Mode A under the flat-in-redshift prior, but its baseline-volumetric-prior mass is below 5% of Mode A’s, so reweighted samples are essentially blind to it. Diagnostics such as an effective-sample-size ratio under the reweighted weights, or a direct overlay of the  $d_L$  marginals, can flag such cases before the reweighted summary is reported.

### 6.3 Implications for future bright sirens

The practical consequence is that direct prior sampling should be used for bright-siren  $H_0$  analyses whenever the target prior differs non-trivially in support from the baseline, rather than being used only as a cross-check. The runtime budget on a single A100 GPU now permits such reruns routinely (Section 5); the live-point and sampler-robustness studies show that the science-quality configuration sits in the converged regime, so the marginal cost of a prior-sensitivity rerun is the marginal cost of one  $\sim 15$  min run. Live-point counts up to  $10^5$  are accessible on the same hardware, which means that prior-sensitivity studies can in principle be combined with population-level





**Figure 6.** GW170817 wall-clock scaling. Heterodyned and unheterodyned wall-clock as a function of  $n_{\text{live}}$  for the IMRPhenomD\_NRTidalv2 family on default-mass and LVK-bounds prior sets, with the TaylorF2 unheterodyned point at  $n_{\text{live}} = 1500$  for context. The heterodyned LVK-bounds series has log-log gradient  $\approx 0.60$  over the plotted range; matched unheterodyned points are  $31\times/51\times/68\times$  slower at  $n_{\text{live}} = 500/1500/2500$ .

analyses without a substantial increase in end-to-end wall-clock time. This is directly relevant to the third-generation detector era (Hu & Veitch 2025), where the detection rate of compact-binary mergers will make repeated, controlled posterior reanalyses the norm rather than the exception.

#### 6.4 Limitations and open items

Several limitations should be acknowledged. First, no waveform in our JAX inventory simultaneously includes precession and tides; the locked primary is therefore the aligned-spin tidal IMRPhenomXAS\_NRTidalv3, and the precession contribution is bracketed inline by the IMRPhenomPv2 (no tides) cross-check rather than by a precessing tidal model. A like-for-like reproduction of the LVK IMRPhenomPv2\_NRTidal analysis would require porting that waveform to JAX. Second, the host-galaxy association with NGC 4993 and the peculiar-velocity inputs are taken directly from Abbott et al. (2017b); we have varied  $\sigma_{v_p}$  but not  $\langle v_p \rangle$ . Third, we do not present a matched PARALLEL\_BILBY benchmark; the performance claim is therefore explicitly restricted to single-A100 wall-clock runtime on the current pipeline. Fourth, the  $d_L$ - $\iota$  bimodality is characterised at the level of local evidences and mode-isolated marginals; we do not at present propagate Mode B into a population analysis or combine it with an electromagnetic likelihood for the host inclination.

## 7 CONCLUSIONS

We have used a GPU-native nested-sampling pipeline to revisit the GW170817 bright-siren  $H_0$  measurement of Abbott et al. (2017b) with the locked primary aligned-spin tidal waveform IMRPhenomXAS\_NRTidalv3, focused on whether the post-hoc prior-reweighting step in that analysis recovers the answer that direct sampling under the target prior would give. On the LVK-bounds prior set, IMRPhenomXAS\_NRTidalv3 yields  $H_0^{\text{MAP}} = 72.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$  with sample-derived 68 per cent HPD  $[63.1, 88.5] \text{ km s}^{-1} \text{ Mpc}^{-1}$ ; IMRPhenomD\_NRTidalv2 and TaylorF2 give consistent values within their HPDs, and the GWTC-1 IMRPhenomPv2\_NRTidal posterior mapped through this work’s standard-siren likelihood peaks at  $67.1 \text{ km s}^{-1} \text{ Mpc}^{-1}$  within the same envelope. The four-variant prior-sensitivity sweep on the default-mass full-sky prior set has baseline  $H_0^{\text{MAP}} = 73.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ,  $P(H_0 > 120 \text{ km s}^{-1} \text{ Mpc}^{-1}) = 0.017$ ; switching the distance prior to flat-in-redshift by direct sampling raises that tail probability to 0.159, while post-hoc reweighting captures only 0.041 – approximately 17 % of the shift. The four IMRPhenomXAS\_NRTidalv3 evidence values agree to  $\Delta \ln Z \lesssim 1.8$ , so the data do not prefer one distance prior over another: the  $H_0$  shift is a prior effect, not a data-driven update. The mechanism is the  $d_L$ - $\iota$  bimodality: Mode A ( $d_L \in [30, 75] \text{ Mpc}$ ,  $H_0^{\text{MAP}} = 74.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ) and Mode B ( $d_L \in [10, 30] \text{ Mpc}$ ,  $H_0^{\text{MAP}} = 110.8 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ) carry comparable local evidence af-

ter prior-volume normalisation ( $\ln \mathcal{B}_{B/A} = -0.66$ ); reweighting cannot recover Mode B because the baseline volumetric prior assigns it negligible mass.

The heterodyned likelihood gives wall-clock speedups of 31 $\times$ , 51 $\times$ , and 68 $\times$  over the unheterodyned reference at  $n_{\text{live}} = 500$ , 1500, 2500 on a single A100 GPU. A matched-pair sky-prior sweep at  $n_{\text{live}} = 5000$  confirms that imposing a  $\pm 0.05$  rad patch around NGC 4993 changes wall-clock by only 0.86 $\times$  for the heterodyned IMRPhenomD\_NRTidalv2 sampler, 0.98 $\times$  for unheterodyned IMRPhenomD\_NRTidalv2, and 1.44 $\times$  for unheterodyned TaylorF2, so the speedup is not driven by a tight sky prior. The scientific value of fast nested sampling for bright-siren cosmology is that direct prior reruns, multi-waveform cross-checks, and full sampler-robustness studies all become routine, replacing post-hoc reweighting and isolated cross-checks as the default robustness tool without a substantial change in the computational budget.

## ACKNOWLEDGEMENTS

TODO: add supervisor, research-group, compute-resource (Google Cloud A100), and funding acknowledgements; acknowledge GWOSC (TODO: insert appropriate GWOSC data-release citation) and the LIGO–Virgo–KAGRA Collaboration for the public strain data and reference posteriors used here.

## DATA AVAILABILITY

This paper uses publicly available GW170817 and GW150914 strain data and GWTC-1/GWTC-2.1 reference posteriors from the Gravitational Wave Open Science Center (GWOSC). All derived posterior-sample CSV files, summary tables, scaling logs, and figure-generation scripts are distributed with the project repository at [TODO:insertpermanentrepositoryURL/DOI](#), with provenance recorded in `paper_knowledge_base/result_link_index.md` and the scaling-study summary CSV. TODO: confirm GWOSC release identifiers for the GW170817 and GW150914 strain segments before submission.

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